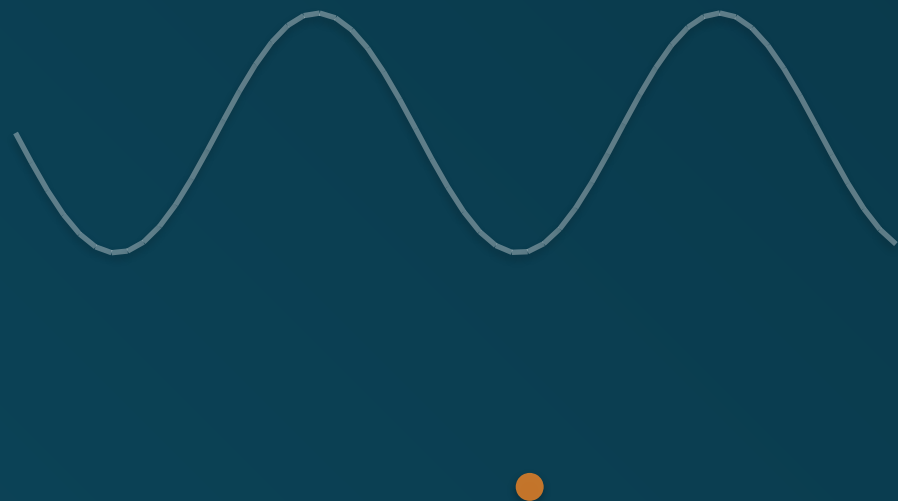


TOPIC 16

Thermodynamics

热力学



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

16.1 Internal energy 内能

- Internal energy is determined by the state of the system 内能由系统状态决定
- Internal energy = sum of a random distribution of kinetic and potential energies of the molecules 分子动能 + 势能总和
- A rise in temperature relates to an increase in internal energy 温度升高 $\Rightarrow$ 内能增加

16.2 The first law of thermodynamics 热力学第一定律

- Recall and use $w = p\Delta V$ for work done at constant pressure 定压做功公式
- Distinguish work done BY the gas and work done ON the gas 区分气体对外 / 外界对气体做功
- Recall and use the first law $\Delta U = q + w$ 热力学第一定律

Internal Energy

内能

Internal energy U : the sum of a **random distribution of kinetic and potential energies** associated with the molecules of a system. It is determined by the **state** of the system.

系统内分子无规则动能与势能的总和，由系统状态决定。

- **Rise in temperature** $\Rightarrow$ increase in internal energy 温度升高 $\Rightarrow$ 内能增加（分子平均动能增大）
- During a **phase change**, temperature is constant: KE unchanged, PE increases 相变时动能不变、势能增大
- For an **ideal gas**: no intermolecular forces $\Rightarrow$ no PE $\Rightarrow U$ = total kinetic energy of molecules 理想气体内能 = 分子动能总和
- U is a **state function**: depends only on the state, not on how the system got there 内能只取决于状态

Work Done by a Gas

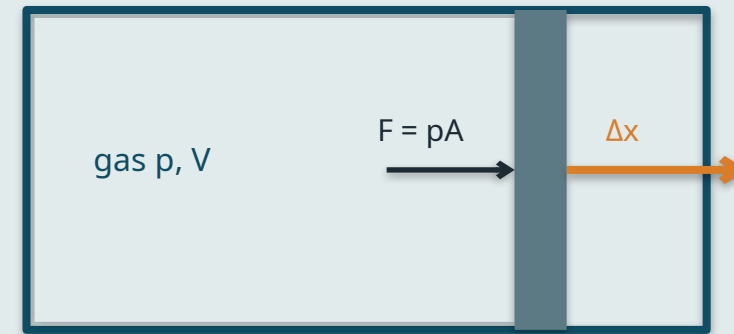
气体做功 $w = p\Delta V$

- Gas at pressure p pushes piston of area A out by Δx :
- force ; work 推导

$$w = p \Delta V$$

constant pressure 定压过程

- **Gas expands** ($\Delta V > 0$): work done **BY** the gas 气体对外做功
- **Gas is compressed** ($\Delta V < 0$): work done **ON** the gas 外界对气体做功
- Units: p in Pa, V in m^3 , w in J 统一 SI 单位

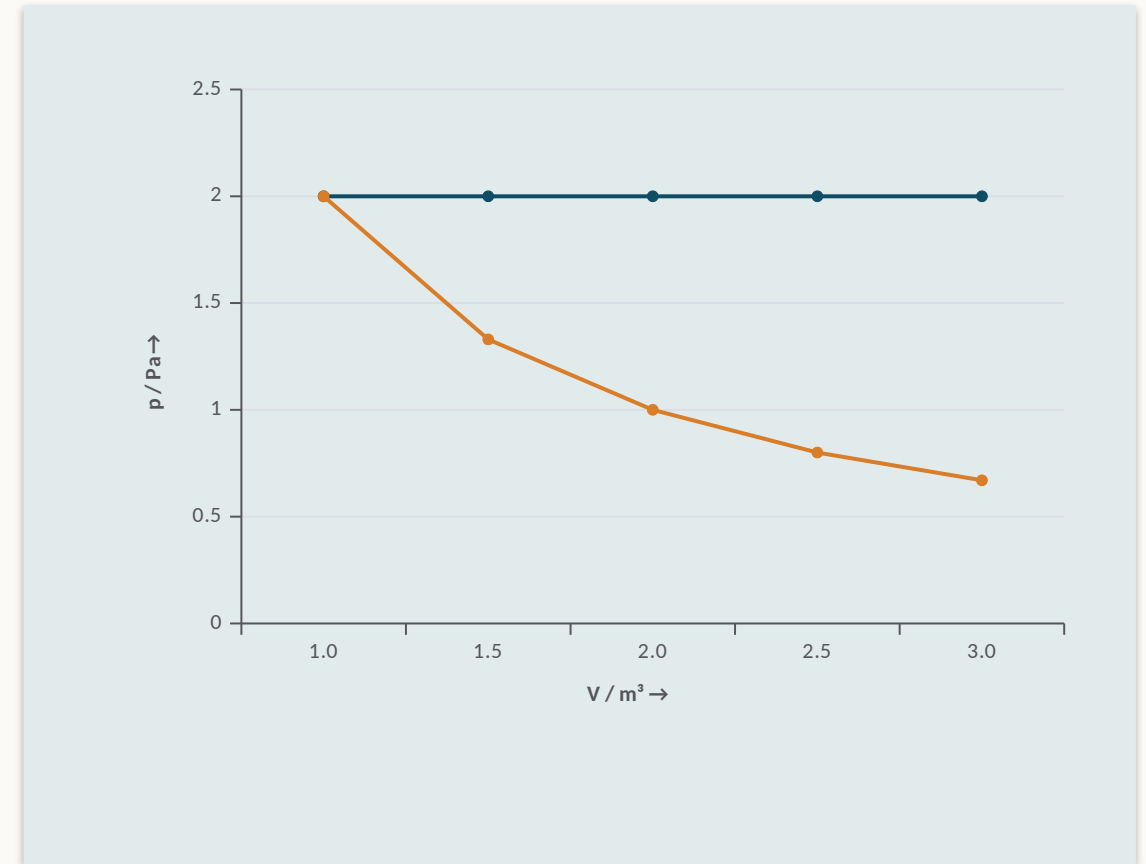


Expansion at constant pressure 定压膨胀: 气体推动活塞对外做功

Work from p-V Diagrams

p-V 图与做功

- **Area under a p-V curve = work done** during the change 曲线下面积 = 做功
- Constant pressure: rectangular area 等压: 矩形面积
- Changing pressure: split the area into thin strips and sum 变压力: 分割求和 (数值估算)
- **Isochoric** ($\Delta V = 0$): no area $\Rightarrow$ no work 等容不做功
- A closed cycle: net work = area enclosed by the loop 循环净功 = 闭合面积



Area under curve = work done 曲线下面积即功

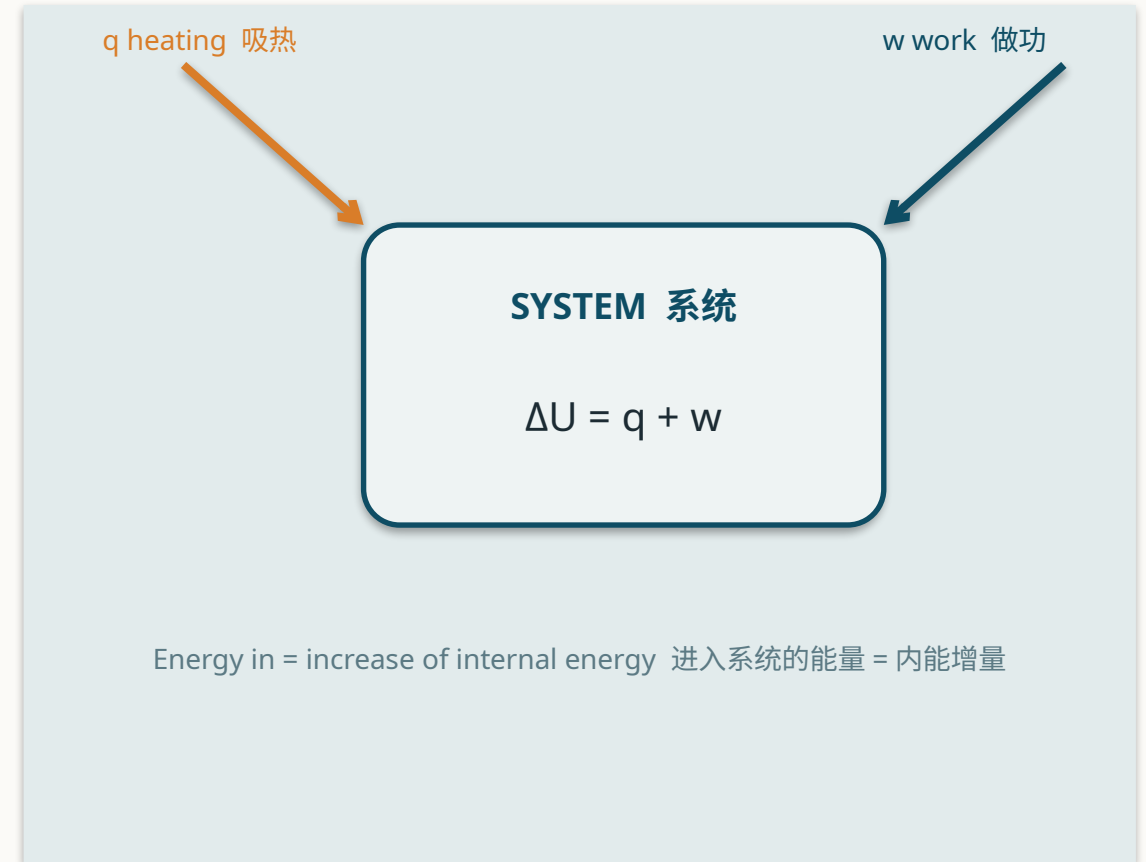
The First Law of Thermodynamics

热力学第一定律

$$\Delta U = q + w$$

能量守恒在热力学中的表达

- : increase in internal energy 内能增加量
- : energy transferred **TO** the system by heating 系统吸热为正
- : work done **ON** the system 对系统做功为正
- It is the **law of conservation of energy** applied to thermal systems 本质是能量守恒



Applying the First Law

典型过程分析

Process 过程	Condition 条件	Consequence 结论	Physics 物理意义
Isochoric 等容	$\Delta V = 0$	$w = 0 \Rightarrow \Delta U = q$	All heat goes into internal energy 吸热全变内能
Isobaric 等压	$p = \text{const}$	$w = p\Delta V; \Delta U = q + w$	Heat = ΔU + expansion work 吸热部分对外做功
Isothermal 等温	$T = \text{const}$	$\Delta U = 0 \Rightarrow q = -w$	Heat in = work out (ideal gas) 吸热全部对外做功
Adiabatic 绝热	$q = 0$	$\Delta U = w$	Work changes internal energy 做功直接改变内能

• Sign convention 符号约定 : $q > 0$ 系统吸热; $w > 0$ 对系统做功; $\Delta U > 0$ 内能增加

Worked Example 16A

例题解析

WORKED EXAMPLE 16A

A sample of gas has a volume of 750 cm^3 . The gas expands at a constant pressure of until its volume is 900 cm^3 . Calculate the work done by the gas during the expansion.

$\text{cm}^3 \rightarrow \text{m}^3$ 要 $\times 10^{-6}$; 膨胀即气体对外做功。

SOLUTION 解答

Step 1 Convert volumes: .

Step 2 Constant pressure $\Rightarrow$.

Step 3

Answer (done by the gas 气体对外做功)

Worked Example 16B

例题解析

WORKED EXAMPLE 16B

250 J of thermal energy is added to a system, and the system does 100 J of work. Find the change in internal energy of the system.

关键是符号：系统对外做功 $\Rightarrow$ w 取负。

SOLUTION 解答

Step 1 First law: .

Step 2 (heat supplied to the system 系统吸热).

Step 3 The system does work $\Rightarrow$ work done ON the system is negative: .

Step 4

Answer (**increase** 内能增加)

Worked Example 16C

例题解析

WORKED EXAMPLE 16C

The specific latent heat of vaporisation of a liquid is 2300 J g^{-1} . When 1.0 g of the liquid evaporates, its volume increases by 1600 cm^3 against atmospheric pressure. Determine (a) the work done against the atmosphere; (b) the increase in internal energy.

汽化潜热一部分变为内能，一部分用于对抗大气膨胀做功。

SOLUTION 解答

Step 1 (a) .

Step 2 (b) Heat supplied .

Step 3 First law with (work done by the gas):

Step 4

Answer (a) **160 J** (b)

Class Practice

随堂练习

Q1 Air in a tyre occupies . When the tyre bursts, the air expands to against atmospheric pressure . Calculate the work done against the atmosphere.

Q2 A gas is compressed so that 500 J of work is done on it, while 200 J of heat escapes to the surroundings. Find the change in internal energy. 压缩 + 放热求 ΔU 。

Q3 Explain why $\Delta U = 0$ in any isothermal change of an ideal gas. In such a change, 350 J of heat is added to the system — how much work is done, and by whom? 等温过程分析。

Answers Q1 Q2 Q3 T 不变 $\Rightarrow$ 分子动能不变 $\Rightarrow \Delta U = 0$; 系统对外做功 350 J

Summary

本章小结

KEY FORMULAS 核心公式

$$U = \sum \dot{Q}$$

internal energy 内能

$$w = p \Delta V$$

work at const. p

$$\Delta U = q + w$$

first law 第一定律

$$q > 0: \text{heat in}; w > 0: \text{work on}$$

sign convention 符号约定

KEY IDEAS 核心思想

- Internal energy depends only on the state of the system 内能是状态量。
- Area under a p-V curve gives the work done p-V 面积即功。
- Isothermal ideal gas: $\Delta U = 0$; adiabatic: $\Delta U = w$. 两个特例常考。

Exam Tips

备考提示

- Read carefully: "work done **BY** the gas" vs "work done **ON** the gas" — they differ by a sign. 分清对外 / 对内做功。
- ; p in Pa, V in m^3 ($\text{cm}^3 \times 10^{-6}$). 单位统一。
- State the sign convention before substituting into — examiners award a mark for it. 先声明符号约定。
- Expect combined questions with Topic 15: use to find states, then the first law for energies. 常与理想气体综合出题。